

WICHITA JABARA, KS, USA

WMO: 724504

Lat: 37.746N

Lon: 97.221W

Elev: 1421

StdP: 13.96

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 03974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	11.4	-2.9	5.0	10.5	1.1	6.1	14.1	31.9	38.3	28.4	39.1	10.2	350	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	99.5	73.9	96.6	74.2	92.8	74.0	77.8	90.5	76.7	89.6	75.6	88.4	13.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.2	134.8	84.1	73.0	129.1	82.7	72.1	125.4	81.7	42.3	90.7	41.1	89.6	40.1	88.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.7	25.0	22.6	0.5	103.6	5.8	3.4	-3.7	106.1	-7.1	108.0	-10.3	109.9	-14.6	112.4		
			0.0	80.3	5.8	1.4	-4.1	81.3	-7.5	82.1	-10.7	82.9	-14.9	83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	33.2	37.2	46.8	56.6	66.3	75.6	80.6	78.9	71.0	58.7	46.2	35.1
	DBStd	18.98	11.40	11.73	11.02	9.42	7.92	6.00	5.96	6.12	7.48	9.42	10.06	10.19
	HDD50	1822	526	379	190	43	1	0	0	0	0	30	185	468
	HDD65	4453	985	778	570	277	84	4	0	0	28	236	565	926
	CDD50	4483	6	20	89	241	507	769	948	895	629	301	71	7
	CDD65	1642	0	0	5	25	124	323	484	431	206	42	2	0
	CDH74	17712	0	2	46	272	1098	3306	5748	4807	2024	393	16	0
	CDH80	8192	0	0	6	59	347	1417	3019	2419	819	105	1	0
Wind	WSAvg	10.9	11.2	11.4	12.5	13.1	11.2	10.6	9.5	8.9	9.7	10.6	11.1	10.9
Precipitation	PrecAvg	35.3	0.7	1.3	2.4	3.2	5.2	5.7	4.1	4.0	3.1	2.8	1.7	1.1
	PrecMax	50.9	2.6	4.1	4.7	7.4	11.1	10.9	9.1	9.1	10.3	8.1	5.4	3.4
	PrecMin	24.8	0.1	0.0	0.4	0.6	1.6	2.0	0.5	0.4	0.6	0.6	0.0	0.1
	PrecStd	7.0	0.6	1.0	1.3	1.7	2.5	2.4	2.2	2.2	2.4	1.7	1.6	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	72.4	79.3	85.9	91.5	98.1	104.2	103.6	97.4	88.1	75.6	65.7
		MCWB	51.0	54.6	60.0	64.8	69.9	73.7	73.7	73.5	71.9	67.8	59.0	55.0
	2%	DB	60.7	66.1	73.5	81.1	87.7	94.0	100.1	99.3	91.4	82.4	71.9	60.5
		MCWB	47.7	51.7	59.2	63.2	70.5	74.4	74.2	73.6	71.9	65.7	58.4	49.6
	5%	DB	55.3	60.9	69.5	76.9	83.8	90.9	97.1	96.0	88.4	78.6	66.3	55.0
		MCWB	45.0	49.5	56.6	61.5	68.8	73.8	74.7	74.1	71.6	63.6	55.3	46.9
	10%	DB	50.5	55.2	64.5	72.7	80.6	88.1	93.1	91.4	84.4	73.7	62.5	50.8
		MCWB	42.2	46.1	54.0	59.7	68.0	73.1	74.8	73.9	69.5	62.7	53.4	43.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	58.9	65.8	68.8	74.8	78.9	79.5	79.1	76.5	72.7	64.0	59.3
		MCDB	59.8	64.5	74.8	80.1	85.0	90.8	92.4	90.7	88.3	81.3	71.2	64.3
	2%	WB	50.0	54.9	62.4	66.8	73.0	77.2	77.8	77.4	74.5	70.1	60.1	52.2
		MCDB	58.6	62.4	68.9	75.8	83.1	89.1	91.0	90.5	86.3	77.5	67.1	56.3
	5%	WB	45.9	50.8	59.3	64.5	71.3	75.7	76.8	76.2	73.1	67.4	57.8	47.6
		MCDB	54.1	60.9	66.5	73.1	80.4	87.2	90.2	89.1	84.6	74.2	64.5	54.1
	10%	WB	42.4	46.1	55.1	61.9	69.5	74.3	75.7	75.0	71.6	64.2	54.6	44.0
		MCDB	50.6	54.7	63.9	71.1	78.1	85.0	89.4	87.5	81.8	71.5	61.5	50.6
Mean Daily Temperature Range	5% DB	MDBR	20.1	20.8	22.1	22.0	20.6	20.2	21.3	21.2	21.8	22.0	21.1	18.7
		MCDBR	27.7	28.8	28.3	27.0	24.3	22.6	25.3	25.8	24.5	26.6	26.3	25.5
		MCWBR	17.2	17.9	16.2	14.9	11.4	8.7	7.1	7.6	9.6	12.9	15.1	16.9
	5% WB	MCDBR	26.2	26.6	22.8	23.3	20.9	20.5	21.5	21.6	21.0	20.6	21.6	23.4
		MCWBR	17.5	17.9	15.4	14.9	11.6	9.5	8.3	8.5	9.2	12.4	14.9	17.5
Clear-Sky Solar Irradiance	taub	0.286	0.298	0.332	0.373	0.407	0.412	0.420	0.424	0.379	0.336	0.297	0.280	
	taud	2.501	2.475	2.411	2.319	2.297	2.312	2.331	2.276	2.377	2.486	2.526	2.525	
	Ebn at Noon	288	298	296	287	278	276	272	269	276	280	281	283	
	Edh at Noon	25	29	34	40	42	41	40	41	35	29	24	23	
All-Sky Solar Radiation	RadAvg	795	1026	1340	1686	1890	2158	2125	1905	1582	1171	862	681	
	RadStd	79	144	91	111	165	155	132	106	113	132	85	65	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.67	+0.85	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page